

An SAR Target Detector Based on Gradient Harmonized Mechanism and Attention Mechanism

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Abstract—In this letter, a target detector based on gradient harmonized mechanism (GHM) and attention mechanism is proposed to realize synthetic aperture radar (SAR) target detection in complex scenes. Considering the imbalance of positive and negative examples in SAR target detection, we use RefineDet as our backbone network. RefineDet can mitigate this imbalance problem by introducing the idea of two-step classification and regression into the one-stage detector. However, RefineDet only selects a part of examples for training and does not make full use of the information of all examples. Therefore, we apply GHM to the classification loss function of RefineDet, so that the network can make full use of all examples and increase the weights of hard examples adaptively in the loss function to reduce the false alarms and the missing alarms. In addition, to achieve a better detection performance in SAR images with complex scenes, a multiscale feature attention module (MFAM) is embedded into the network. By applying channel and spatial attention mechanisms to the multiscale feature maps, the MFAM can highlight the significant information and suppress the interference caused by clutter. The extensive experimental results based on the measured SAR dataset verify the effectiveness of the proposed method.

Index Terms—Gradient harmonized mechanism (GHM), multiscale feature attention module (MFAM), synthetic aperture radar (SAR), target detection.

I. INTRODUCTION

AS A high-resolution active microwave imaging radar, synthetic aperture radar (SAR) has the advantages of all-weather and all-day compared with the optical imaging systems. With the rapid development of SAR related technologies, it is no longer difficult to acquire high-resolution SAR images, which promotes the application of SAR target detection technology. Researchers have extensively studied the traditional constant false alarm rate (CFAR) [1] methods. The two-parameter CFAR detector [2] which assumes the clutter obeying the Gaussian distribution has been widely used in 2-D SAR images. The detection methods based on CFAR perform well in some simple scenes. However, for the urban areas and the land-sea boundary areas with much complex clutter, the performance of the CFAR methods will be degraded.

In recent years, target detection methods based on deep learning have been developed rapidly. Both the two-stage

detector (e.g., Faster R-CNN [3]) and the one-stage detector (e.g., single-shot multibox detector (SSD) [4]) have attracted extensive attention from researchers. Some methods [5]–[7] apply the one-stage target detector and two-stage target detector, respectively, to SAR target detection. These methods achieved good detection results. Most two-stage and one-stage target detection methods are anchor-based methods. The model generates a large number of anchors with different sizes and aspect ratios and then performs classification and bounding box regression on these anchors.

In the anchor-based SAR target detection network, a large number of anchor boxes are needed to ensure sufficient overlap with most ground-truth boxes in SAR images. However, the target area in SAR images takes up a small proportion of the whole image. In this case, a large number of anchor boxes are the negative examples, which leads to the imbalance of positive and negative examples. Some methods [8], [9] combined the focal loss [10] in the network to mitigate it. However, these methods introduce extra hyperparameters that require a cumbersome cross-validation process to determine. In the field of optical image target detection, RefineDet [11] introduces the idea of two-step classification and regression into the one-stage detector, so that the network can filter out most negative anchors in the second-step classification. RefineDet can mitigate this imbalance problem to some extent without introducing any extra hyperparameters and has the advantage of high efficiency meanwhile.

According to the above analysis, we propose an SAR target detector based on gradient harmonized mechanism (GHM) and attention mechanism. Although RefineDet can mitigate the imbalance of positive and negative examples, it still has two shortcomings. First, RefineDet only selects some examples for training in each iteration. It does not make full use of the information of all examples. Second, for the hard examples that are difficult to distinguish between the target and the clutter and the easy examples, RefineDet pays the same attention to them. These problems make the network generate more false alarms and missing alarms in the SAR images with complex scenes. Considering these problems, we apply GHM [12] to the classification loss of RefineDet. It enables the network to make full use of all examples and increases the weights of hard examples adaptively in the loss function without introducing any extra hyperparameters. Therefore, our network can focus more on the hard examples to reduce the false alarms and missing alarms. To the best of our knowledge, this is the first application of GHM in SAR target detection. In addition, we propose a novel multiscale feature attention module (MFAM). By applying the channel and spatial

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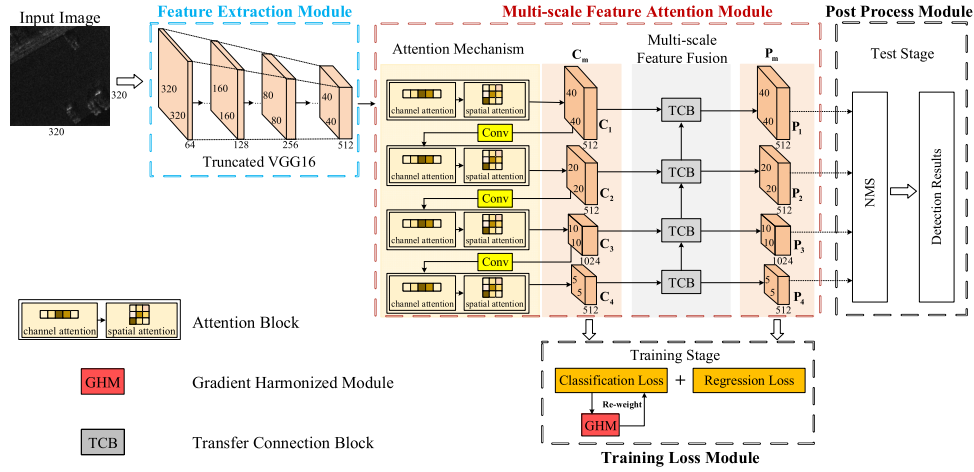


Fig. 1. Architecture of the proposed SAR target detection network.

attention mechanisms to multiscale feature maps, the MFAM can highlight the significant information and suppress the interference caused by complex surroundings. The extensive experimental results based on the measured miniSAR [13] dataset and AIR-SARShip-1.0 [14] dataset demonstrate the effectiveness of the proposed method.

II. PROPOSED TARGET DETECTION METHOD

Fig. 1 shows the architecture of the proposed SAR target detection network. The network consists of four components: feature extraction module (FEM), MFAM, training loss module in the training stage, and postprocess module in the test stage. The FEM is a truncated VGG16, which is used to extract deep features of the input SAR images. The MFAM is used to obtain the multiscale feature maps C_m and P_m , in which $m \in [1, 2, 3, 4]$ represents the m th scale feature map of MFAM. In each scale of MFAM, an attention block including a channel attention module and a spatial attention module is adopted to highlight the significant features in the channel and spatial dimensions of C_m . Then the highlighted features are converted into P_m by the transfer connection block (TCB). The attention block can refine the feature maps C_m and P_m adaptively, which benefits to improve the detection performance in SAR images with complex scenes. Same as RefineDet, our network contains two-step classification and regression. Specifically, the network first gets the results of the classification and rough regression processes of the first step based on C_m . According to the classification results of the first step, a positive confidence score indicating the presence of targets will be obtained for each anchor. Then these anchors with positive confidence scores larger than a preset threshold θ (i.e., set $\theta = 0.01$ empirically) will participate in the classification and fine regression processes of the second step based on P_m . As for the second-step classification, the first-step classification plays a role in filtering negative examples. In this way, the number of negative examples reduced significantly, which mitigates the imbalance of positive and negative examples. In the training stage, the output results of both the steps are compared with the ground truth. The GHM is applied to the classification loss function, so that the network can

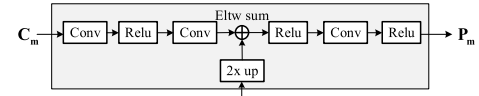


Fig. 2. Architecture of the TCB.

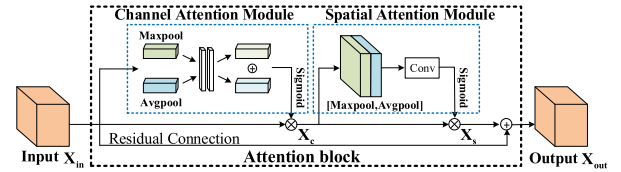


Fig. 3. Architecture of the attention block.

make full use of all examples and increase the weights of hard examples adaptively. In the test stage, only the output results of the second step are used as the predicted results of the network. The postprocess module obtains the final detection results after filtering the predicted results via nonmaximum suppression (NMS).

In the following sections, the structures of MFAM and training loss module are introduced in detail.

A. MFAM

As shown in Fig. 1, the MFAM contains feature maps of four scales, the sizes of which are 40×40 , 20×20 , 10×10 , and 5×5 , respectively. In each scale, we introduce the convolutional block attention module [15] as our attention block to refine the feature maps C_m and P_m . Different from other multiscale attention methods [5], [6], since we embed the attention block between C_m , the highlighted features of different scales can be fully integrated in a way that combines top-down and bottom-up in the MFAM. Note that the TCB can integrate the large-scale context from C_m and convert it into P_m . Specifically, it first upsamples the high-level feature and sums it with the lower level feature in an elementwise way. Then, a convolution layer is added after summation to generate P_m . The detailed architecture of the TCB is shown in Fig. 2.

The detailed architecture of the attention block is shown in Fig. 3. Given the input feature map $X_{in} \in \mathbb{R}^{C \times H \times W}$,

it will pass through the channel attention module and the spatial attention module to obtain attention descriptors in turn. In the channel attention module, the average pooling and max pooling are first adopted along the spatial dimension. Then a shared multiple-layer perceptron (MLP) is used to produce the channel attention descriptor $\mathbf{A}_c \in \mathbb{R}^{C \times 1 \times 1}$. The computation process of the channel attention module is summarized as follows:

$$\mathbf{A}_c = \sigma(\text{MLP}(\text{Avg Pool}(\mathbf{X}_{\text{in}})) + \text{MLP}(\text{Max Pool}(\mathbf{X}_{\text{in}}))) \quad (1)$$

where σ refers to the sigmoid function. Then $\mathbf{X}_{\text{in}}$ is multiplied by $\mathbf{A}_c$ to get the channelwise refined feature maps $\mathbf{X}_c$

$$\mathbf{X}_c = \mathbf{A}_c \otimes \mathbf{X}_{\text{in}} \quad (2)$$

where $\otimes$ denotes elementwise multiplication. During multiplication, the attention values of $\mathbf{A}_c$ are broadcasted along the spatial dimension.

Same as the channel attention module, the spatial attention module first adopts the average pooling operation and max-pooling operation but along the channel dimension, and it concatenates the results. Then a 7×7 convolution layer is applied to generate the spatial attention descriptor $\mathbf{A}_s \in \mathbb{R}^{1 \times H \times W}$. It is computed as follows:

$$\mathbf{A}_s = \sigma(\text{Conv}_{7 \times 7}([\text{Avg Pool}(\mathbf{X}_c); \text{Max Pool}(\mathbf{X}_c)])) \quad (3)$$

We can get the spatialwise refined feature maps $\mathbf{X}_s$ by multiplying $\mathbf{A}_s$ and $\mathbf{X}_c$

$$\mathbf{X}_s = \mathbf{A}_s \otimes \mathbf{X}_c \quad (4)$$

The attention values of $\mathbf{A}_s$ are also broadcasted but along the channel dimension. Then the final refined output feature map $\mathbf{X}_{\text{out}} \in \mathbb{R}^{C \times H \times W}$ is obtained via residual connection

$$\mathbf{X}_{\text{out}} = \mathbf{X}_s + \mathbf{X}_{\text{in}} \quad (5)$$

B. Training Loss Module

RefineDet only selects a part of negative examples with top loss values for training in the classification processes of both the steps. It does not make full use of all examples. The reasons are as follows: consider the binary cross entropy loss:

$$L_{\text{cross_entropy}} = \begin{cases} -\log(p), & \text{if } p^* = 1 \\ -\log(1-p), & \text{if } p^* = 0 \end{cases} \quad (6)$$

where $p = \text{sigmoid}(x)$ is the probability predicted by the network, x is the output of the network, and p^* is its ground-truth label. The gradient norm g with regard to x is

$$g = \left| \frac{\partial L_{\text{cross_entropy}}}{\partial x} \right| = \begin{cases} 1-p, & \text{if } p^* = 1 \\ p, & \text{if } p^* = 0. \end{cases} \quad (7)$$

The value of g indicates the degree of difficulty of the current example; the larger the value of g , the more difficult it is for the current example to distinguish between the target and the clutter. Fig. 4 shows the distribution of the gradient norm of examples in the RefineDet which has converged. The y-axis denotes the fraction of examples. We can find that the number of negative examples is very large when the gradient norm is

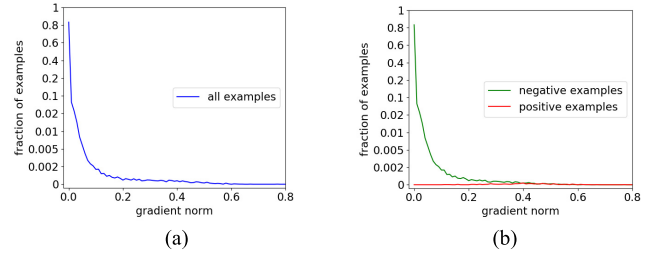


Fig. 4. Distribution of the gradient norm; y-axis denotes the fraction of examples. (a) Distribution of gradient norm of all examples. (b) Distribution of gradient norm of positive and negative examples.

between 0 and 0.1. It means that the network will generate a large number of easy negative examples when we use all examples, and the total contribution of the easy examples can overwhelm the contribution of the minority of hard examples. This makes the training process of the network inefficient, and even the network cannot converge. However, the network can use all examples by applying GHM to the classification loss function. The details of GHM are as follows.

According to the characteristics of a large number of easy negative examples and a small number of hard examples during network training, we introduce the gradient density function to help adjust the weights of examples adaptively. It is formulated as follows:

$$d(g) = \frac{1}{l_\varepsilon(g)} \sum_{n=1}^N \delta_\varepsilon(g_n, g) \quad (8)$$

where n is the index of anchor in a minibatch, N is the total number of examples, and g_n is the gradient norm of the n th example. And

$$l_\varepsilon(g) = \min\left(g + \frac{\varepsilon}{2}, 1\right) - \max\left(g - \frac{\varepsilon}{2}, 0\right) \quad (9)$$

$$\delta_\varepsilon(g_n, g) = \begin{cases} 1, & \text{if } g - \frac{\varepsilon}{2} \leq g_n < g + \frac{\varepsilon}{2} \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

where $d(g)$ denotes the number of examples whose gradient norm is within the region centered at g with a length of ε and normalized by the valid length of the region. We can set different weights for examples in different gradient norm regions according to $d(g)$

$$\beta_i = \frac{1}{d(g_i)/N} = \frac{N}{d(g_i)} \quad (11)$$

where β_i is the weight of the i th example, and it is inversely proportional to $d(g_i)$. The reformulated classification loss function of the network is given by the following equation:

$$L_{\text{cls}} = \frac{1}{N} \sum_{i=1}^N \beta_i L_{\text{cross_entropy}}(p_i, p_i^*). \quad (12)$$

By applying GHM, the easy negative examples with large gradient density will be down-weighted, i.e., the β of easy negative examples is relatively small, and the hard examples with small gradient density will be up-weighted, i.e., β of hard examples is relatively large. Therefore, the network can use all examples and pay more attention to the hard examples that

are difficult to distinguish between the target and the clutter. We use Smooth L1 loss as our regression loss. The total loss function of the network is given by the following equation:

$$L(p_i^1, p_i^2, t_i^1, t_i^2) = L_{\text{cls}}(p_i^1, p_i^*) + p_i^* L_{\text{reg}}(t_i^1, t_i^*) + L_{\text{cls}}(p_i^2, p_i^*) + p_i^* L_{\text{reg}}(t_i^2, t_i^*) \quad (13)$$

where p_i^1 and t_i^1 are the output results of the first step, which denote the positive confidence score and the predicted coordinates of the bounding box, respectively, of the i th example. Similarly, p_i^2 and t_i^2 are the output results of the second step.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset Description and Experimental Settings

The proposed method is verified with two measured SAR datasets: the miniSAR dataset with vehicles are the targets of interest and the AIR-SARShip-1.0 dataset with ships are the targets of interest. For the miniSAR dataset, nine SAR images of large scenes are selected for the experiment, seven of which for training and two for test. The AIR-SARShip-1.0 dataset comprises 31 SAR images of large scenes, 21 of which are for training and ten for test. The training hyperparameters are set as follows: we train our methods using stochastic gradient descent (SGD) with the momentum of 0.9 and weight decay of 0.0005. The learning rate is set as 0.001. The length of the gradient norm region ε in GHM is set as 0.1. The experiments are implemented with Pytorch framework, using a personal computer with Intel Xeon E5-2630 v4 CPU and NVIDIA GeForce GTX 1080Ti GPU on a Ubuntu 16.04 Linux system.

B. Evaluation Criteria

We use precision, recall, and F1-score to evaluate different detection methods quantitatively. They are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (14)$$

$$\text{Recall} = \frac{\text{TP}}{\text{NP}} \quad (15)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

where TP is the number of correctly detected targets, FP is the number of false alarms, and NP is the number of ground truth targets. The F1-score, which is the harmonic mean between precision and recall, can comprehensively evaluate the performance of target detection.

C. Ablation Study

In this subsection, we conduct ablation experiments on the miniSAR dataset to verify the effectiveness of MFAM and GHM. The results are shown in Table I, where the third row and the fourth row indicate that only the channel attention module and spatial attention module in MFAM, respectively, are added on the basis of RefineDet. It verifies that both the attention modules are effective. From the fifth to seventh rows, we can conclude that both the MFAM and GHM can improve the detection performance, and the proposed method can achieve the best detection performance using them meanwhile.

TABLE I
EFFECTIVENESS OF MFAM AND GHM

	Precision	Recall	F1-score
RefineDet	0.8960	0.9106	0.9032
RefineDet with channel attention module	0.9024	0.9106	0.9065
RefineDet with spatial attention module	0.9113	0.9187	0.9150
RefineDet with MFAM	0.9262	0.9187	0.9224
RefineDet with GHM	0.9426	0.9187	0.9305
Proposed method	0.9661	0.9268	0.9461

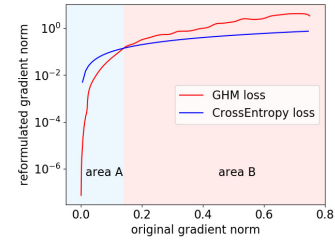


Fig. 5. Reformulated gradient norm of GHM loss and CrossEntropy loss w.r.t. the original gradient norm g .

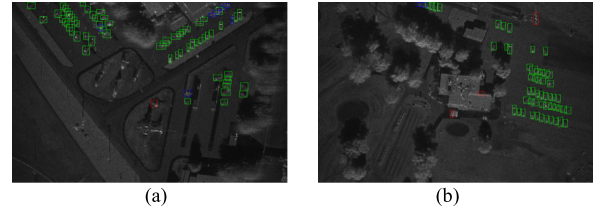


Fig. 6. Target detection results on the miniSAR dataset, where green rectangles represent correctly detected target chips, red rectangles represent false alarms and blue rectangles represent missing alarms. (a) Test image I of miniSAR dataset. (b) Test image II of miniSAR dataset.

Fig. 5 illustrates the reformulated gradient norm of CrossEntropy loss and GHM loss. The x -axis represents the original gradient norm of examples. The y -axis represents the reformulated gradient norm of examples, i.e., the gradient norm after weighting the examples. Note that the y -axis uses log scale to better display the details of GHM loss and CrossEntropy loss. For CrossEntropy loss, the reformulated gradient norm and the original gradient norm are equal, as shown in the blue curve in Fig. 5. For GHM loss, we can find that for the examples in area A with small original gradient norms, their reformulated gradient norms are smaller than their original gradient norms, which means that the gradient contribution of these relatively easy examples is reduced. On the contrary, for the examples in area B, their reformulated gradient norms are larger than their original gradient norms, which means that the gradient contribution of these relatively hard examples is increased. Therefore, our method can focus more on these relatively hard examples, so as to reduce false alarms and missing alarms.

D. Comparison With Other Methods

In this subsection, we compare the proposed method with other state-of-the-art target detection methods on the min-

TABLE II
OVERALL EVALUATION OF DIFFERENT TARGET DETECTION METHODS

	miniSAR dataset			AIR-SARShip-1.0 dataset		
	Precision	Recall	F1-score	Precision	Recall	F1-score
Faster RCNN	0.8370	0.9106	0.8723	0.8190	0.7680	0.7927
SSD	0.8451	0.9106	0.8750	0.7966	0.7600	0.7779
RetinaNet	0.9024	0.8862	0.8942	0.8333	0.7520	0.7906
Faster RCNN with Attention Block	0.8879	0.9024	0.8951	0.8407	0.7840	0.8114
SSD with Attention Block	0.8871	0.9106	0.8987	0.8119	0.7760	0.7936
RetinaNet with Attention Block	0.9300	0.8781	0.9033	0.8440	0.7680	0.8042
RetinaNet with Attention Block and GHM	0.9091	0.9350	0.9218	0.8868	0.7600	0.8185
Proposed method	0.9661	0.9268	0.9461	0.8879	0.8320	0.8591

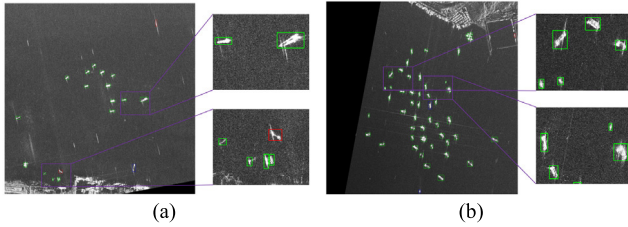


Fig. 7. Target detection results on the AIR-SARShip-1.0 dataset, where green rectangles represent correctly detected target chips, red rectangles represent false alarms and blue rectangles represent missing alarms. (a) Test image I of AIR-SARShip-1.0. (b) Test image II of AIR-SARShip-1.0.

iSAR dataset and AIR-SARShip-1.0 dataset, including Faster R-CNN [3], SSD [4], and RetinaNet [10]. In addition, to be fairer, we also add the attention block and the GHM to these methods and compare their results. Table II shows the quantitative experimental results; it can be seen that the proposed method can achieve the best detection performance for both the miniSAR dataset and AIR-SARShip-1.0 dataset, which proves the effectiveness of the proposed method. Figs. 6 and 7 show the detection results of the proposed method in the SAR images of large complex scenes. We can see that for the urban areas and land-sea boundary areas with much complex clutter, the proposed method still has few missing alarms and false alarms. For some ship targets with different scales, the proposed method can also detect them accurately.

IV. CONCLUSION

In this letter, an SAR target detector based on GHM and attention mechanism is proposed. We use RefineDet as our backbone network to mitigate the imbalance of positive and negative examples. To use all examples in the training process of the network and make the network pay more attention to the hard examples that are difficult to distinguish between target and clutter, we apply GHM to the classification loss function of the network. We propose a novel MFAM to improve the detection performance of SAR images with complex scenes. The MFAM can highlight the significant information and

suppress the interference caused by clutter. The extensive experimental results based on the measured miniSAR dataset and AIR-SARShip-1.0 dataset demonstrate the effectiveness of the proposed method.

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